Certified Feasible-Capacity Operators for Interconnection Queues

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Abstract

Interconnection studies are slow because each request is answered by a separate alternating-current $N-1$ security sweep, so a queue of thousands of requests costs thousands of sweeps and the wait is measured in years. We build three things that make the queue itself the unit of computation. First, a linear security layer whose $N-1$ transfer capacity has an exact closed form. Second, a batched alternating-current $N-1$ bisection that labels every request in a queue in pooled rounds, one round for the whole queue, with each request keeping its own bracket, its own verdict and its own binding contingency. Third, a capacity operator that predicts the derate from the linear capacity to the verified one, trained on the linear layer's own sensitivities, together with a split-conformal certificate and a deferral rule for the requests the certificate cannot admit. On case30 (230 states, 38 studied outages) the pooled labeler returns per-request labels identical to one-request-at-a-time bisection and costs $1.01\times$ less; the operator answers a request in 1.40 ms against 390 power-flow equivalents for the verified label. The certificate attains 91.0% coverage at $1-\epsilon = 10\%$ on held-out states, and the same guarantee does not survive the demand-shifted split, where coverage falls to 66.3% against a 10% target. We report that failure rather than the headline we hoped for: the certificate is a within-distribution claim. The linear anchor it corrects errs in both directions by network, overstating the verified capacity for 95.5% of requests on case30 and for only 3.9% on 118-bus case118, where it is usually conservative instead, so the correction the operator learns changes sign with the network.

1 Introduction

A generator or storage project connects to the grid only after the operator shows that adding it does not overload a branch or push a bus voltage out of band, in the intact network and after any single branch outage. That is an $N-1$ security study per request, tested against every plausible outage at every plausible dispatch of the existing fleet. The queue waiting on those studies dominates project timelines [Rand et al., 2024], and the reason is arithmetic rather than regulatory: a security study is a bisection over load level whose inner step is one power flow per outage, so its cost is (rounds) $\times$ (outages) power flows, and the queue's cost is that times the number of requests.

The computational response has been to make each study cheaper: contingency screening keeps the worst few outages by a linear estimate [Wood & Wollenberg, 1996], learning replaces the power flow with a regressor [Donon et al., 2020, Zamzam & Baker, 2020], and formal methods bound a trained network's outputs [Venzke & Chatzivasileiadis, 2021]. All of this aims at the inner loop. The observation this paper builds on is that the outer loop has a structure the inner loop does not: requests in a queue share a network, an outage set and an operating state, and differ only in the injection each asks for. The batch axis that a contingency sweep already uses for outages can carry requests as well, so a queue costs about what one request costs: the same power flows, one extra request per bisection round.

We build three things. A linear security layer computes the injection-shift-factor matrix of the intact network and of every single-branch outage and returns the largest transfer scale no constraint bounds,

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in closed form; being exact on the linear model, it is not a reliable predictor of the alternating-current capacity, and which way it is wrong depends on the network, so we use it as the physics anchor of a learned operator rather than as an answer. A pooled bisection produces the verified alternating-current labels for a whole queue, the operator predicts the derate from the linear layer's own sensitivities, and a split-conformal step turns its point prediction into a capacity a caller can act on, with a deferral rule for the requests the certificate cannot admit. The reported results are that pooling is a change in cost and not in answer, with labels identical to one-at-a-time bisection at every queue length tested, and that the certificate holds 91.0% coverage against a 10% target on held-out states and across every level evaluated but falls to 66.3% on a demand-shifted split, so the guarantee is a within-distribution claim and the paper reports it as one.

2 Method

The linear security layer, and the closed form it hands us. With B the bus susceptance matrix and C the branch-to-bus incidence weighted by susceptance, the injection shift factors of the intact network are $H = CB^{-1}$ acting on bus injections, and the flow on branch ℓ is $(Hp)_\ell$. For an outage of branch c this becomes $H^{(c)}$, the rank-one update obtained from the branch's outage distribution factors rather than by re-inverting B. An off-nominal tap ratio and a phase shift enter $H^{(c)}$ as they enter B, so a case with off-nominal taps is handled by the same construction. Cross-checked against PANDAPOWER's `makePTDF` [Thurner et al., 2018] on case14, case30, case57, protocol the shift factors agree to 2.7e-15 absolute error, and the closed form with a dense scan over directions to 7.4e-07 relative error, so the layer is exact on the linear model rather than approximate.

For a request that scales a fixed injection pattern u by α, the flow on branch ℓ in state c is $f_\ell^{(c)}(\alpha) = a_\ell^{(c)} + \alpha\, b_\ell^{(c)}$, with $a^{(c)} = H^{(c)} p_0$ at the base injection p_0 and $b^{(c)} = H^{(c)} u$. Requiring $|f_\ell^{(c)}(\alpha)| \le \bar{f}_\ell$ for every state and every monitored branch gives the transfer capacity

$$\alpha^\star \;=\; \min_{c,\ell}\; \min\left(\frac{\bar{f}_\ell - a_\ell^{(c)}}{b_\ell^{(c)}},\; \frac{\bar{f}_\ell + a_\ell^{(c)}}{-b_\ell^{(c)}} \right), \tag{1}$$

taken over the two directions of flow and over states in which $b_\ell^{(c)} \neq 0$. The minimiser is the binding constraint, so the layer returns both the capacity and which outage on which branch set it. Equation 1 is the exact maximiser on the linear model, which is exactly why it is not the answer: it carries no voltage limit and drops the losses, and it exceeds the verified alternating-current capacity for 95.5% of requests even though its branch flows track the alternating-current flows to 4.6% median relative error (16.5% at the 95th percentile). The disagreement is a property of the model, not a defect of the implementation.

Verified labels, one round per queue. The alternating-current capacity of a request is the largest scale at which every outage sweep is secure, and a sweep is one Newton solve per outage. We batch those solves: every outage shares the unknown vector, the specified injections and the bus-type partition, so the only differences are the admittance matrix and the outaged branch, and the stacked linear solve is the entire cost. Bisection returns the largest scale it actually evaluated as secure, so a label can understate the capacity but never overstate it, which is the direction an honest label has to err in. A power flow that fails to converge counts as insecure, because a diverging Newton iteration is what voltage collapse looks like from outside and a study that read it as "no violation found" would admit requests into a system it can no longer solve.

The queue version puts requests on the same batch axis. For D requests the injections become a (D, n_b) array and the sweep runs on $D \times B$ rows, ordered request-major, with the convengence mask evaluated per request. Every request keeps its own bracket $[\ell_d, h_d]$, so a round evaluates only the requests whose bracket is still open, and the number of rounds is set by the slowest request rather than by their sum. The pooling is bounded by a fixed number of stacked systems rather than a fixed number of requests, so peak memory stays put as the queue grows.

The capacity operator. The learning target is the derate $z = \log(\alpha_{\mathrm{AC}} + \epsilon_0) - \log(\alpha^\star + \epsilon_0)$, the log gap between the verified capacity and the linear closed form, with the offset ϵ_0 at 1.00% of the case's median linear capacity so that a request the linear layer calls unbounded still has a finite target. The input is the physics the linear layer already computed, not the raw state: the per-branch sensitivities $b^{(c)}$ of the request, the base-case post-contingency loadings, and the ratio between the binding constraint and the next-tightest one, which is the quantity that decides how much a linear

verdict can be trusted. A branch arm runs graph convolutions on the network with those sensitivities as node and edge features, and a trunk maps the request-level summary to a latent that is combined by inner product, so the operator is a DeepONet whose branch arm reads the linear layer's output rather than a black box on the state [Lu et al., 2021]. The predicted derate multiplies the linear capacity back into a capacity, which makes the linear layer an architectural anchor: at initialisation the operator reproduces the closed form and training only moves it. The same latent feeds a three-class head over which limit binds, thermal, voltage or already-inadmissible, whose outcome is reported in the appendix.

The certificate. Split conformal prediction turns the point prediction into a capacity with a distribution-free guarantee [Vovk et al., 2005, Lei et al., 2018]. On a calibration split the residuals $|z_{\text{pred}} - z_{\text{true}}|$ are collected, the $\lceil (1-\epsilon)(n+1) \rceil$-th smallest is the quantile q, and the certified capacity is the linear capacity derated by $z_{\text{pred}} + q$, so the interval has width $2q$ around the point prediction and the covered claim is $\alpha_{\text{AC}} \geq \alpha^{\star} e^{-(z_{\text{pred}}+q)}$. A request whose certified capacity falls below the scale it asked for is deferred to the exact bisection. The deferral rate is reported rather than hidden, because a certificate that never admits anything is not a screen. Censored labels, meaning brackets the network never bound, are excluded from calibration and counted.

3 Experiments

Protocol. The academic cases ship no conductor ratings: IEEE 14, 57 and 118 carry a placeholder that would make every overload claim vacuous. We therefore define each thermal rating from the network's own reference $N-1$ sweep, taking a branch's rating as its worst unperturbed post-contingency apparent power divided by 0.55, so the unperturbed state sits at a stated $N-1$ margin and two networks are compared at the same utilisation. The voltage limits are two-tier, a normal-state band and a looser contingency band, which is how they are written in practice and the only arrangement in which the reference state is admissible at all. On case30 the ratings have median 27.8 MW, the normal band is $[0.941, 1.020]$ and the contingency band $[0.835, 1.033]$.
A request is a unit-power-factor injection pattern on one to five buses, drawn from candidate sites ranked by load, with weights log-normal across its support. It is evaluated against an operating state drawn from a crossed grid of demand scale $\{0.90, 1.00, 1.10, 1.20\}$ and existing-renewable output scale $\{0.50, 0.75, 1.00\}$ with per-bus log-normal jitter, so two states with the same scalars are still different problems; a shifted split places demand at 1.35 of nominal, outside the training grid. Screen-based contingency sets use the linear layer's own ranking, so the screen cannot peek at the labels, and the capacity it forgoes is measured rather than assumed: the screened capacity is 1.00 of the all-outage capacity on the state sample checked.
The splits are 869 training, 318 calibration and 311 test requests plus 95 shifted requests on case30, and 102 test requests on 118-buscase118 with the top 12 outages screened. States are discarded when the network does not survive its own base case, a property of the state and not of any request in it; that fraction is the paper's first failure number: 9.2% of training states, 2.5% of test states and 60.0% of shifted states are inadmissible before a request is added. We do not engineer that number to zero, because it is what the study has to say about operating margin at the demand levels a queue actually sees.

Result 1: pooling the queue changes nothing about the answer, and its cost benefit is not established at the queue lengths we can afford. Table 1 and Figure 1 report the labelling cost. The claim that survives every check is equivalence. On a 16-request check the pooled and one-at-a-time paths return the same capacity for every request, and over 80 labels of the real dataset every column the dataset stores is identical, including the binding mode, the evaluation counts and the censoring and non-convergence counters. The reason the two agree to floating-point equality rather than merely closely is that pooling does not change the bisection schedule: each round evaluates the same midpoints on the same brackets, so the two paths visit the same sequence of operating points. The equivalence holds at every queue length tested, 4, 8, 16, 32, with a maximum relative difference of 0%.
The cost claim we can defend is narrower than the one we expected: pooling amortises a fixed per-round overhead over the requests in it, so its benefit should grow with queue length, but at the queue lengths this study can afford the measured timings scatter between $0.68\times$ and $1.77\times$ across 4 queue lengths without a clear trend. We report that as a null result rather than picking the favourable end of the scatter, and we keep the pooled path anyway: it is the form a queue-level screen has to take, and its equivalence to the per-request path is what makes either one usable.

What a request costs is better stated in power-flow equivalents than in seconds, because seconds depend on the machine and the equivalents do not, and it is here that the large number in this paper lives. A verified label costs a median of 390 power flows; the operator's screen costs 0.0014 s and no power flow at all, a 390× reduction in the quantity the queue actually pays in, and 636× in wall clock on the machine that ran it. The operator is a screen and not a replacement: the requests it cannot certify are the ones the pool still has to solve.

Result 2: the certificate holds, and the anchor it corrects errs in both directions. Figure 2 reports empirical coverage against the nominal level. At $1 - \epsilon = 10\%$ the certificate attains 91.0% on the test split, against a 10% target and with a binomial standard error of 1.6%, and it tracks the nominal level across the 5 levels evaluated (81.7% at $1 - \epsilon = 80\%$ through 99.7% at 99%). On the shifted split the same procedure attains 66.3%, which is a failure: the violation rate there is 33.7% against a 10% target, and the shortfall is not sampling noise at 95 requests. Exchangeability is what conformal prediction requires and a demand level outside the training grid is exactly what breaks it, so the honest statement is that the certificate is valid on states drawn like the calibration set and is not valid under this shift, and the deferral rule is the mechanism that would have to be made load-bearing where the guarantee lapses.

The comparison that gives those numbers meaning turned out to be the more interesting result, and it is not the one we expected. On case30 the linear closed form exceeds the verified capacity for 95.5% of test requests, which is the rate at which Eq. 1 used as a direct answer would admit a request the network cannot carry. On 118-bus case118 the same quantity is 3.9%, and the median linear capacity is 0.46 of the verified one, so there the linear layer is conservative instead of optimistic and the sign of the correction the operator has to learn is opposite to the small network's. Meshing is one explanation, the differing rating rules another, and the limitations take that confound up; either way the practical consequence is the same, and it is the reason a learned correction is worth having rather than a fixed derate factor: a constant factor chosen on one network has the wrong sign on the other. The certificate is conservative on both, with a median certified-to-verified ratio of 0.90 on case30 and 0.61 on case118, and at $1 - \epsilon = 10\%$ the conformal quantile is negative on the large network (-0.335) because the point prediction already sits below the verified capacity for most requests, which is the price of a distribution-free guarantee rather than a defect.

4 Related work

The linear model, and the shift factors Eq. 1 is built from, are the workhorse of contingency analysis [Wood & Wollenberg, 1996], with verified implementations in MATPOWER and pandapower [Zimmerman et al., 2011, Thurner et al., 2018]. Learning for power systems has largely aimed at the inner solve: learned solvers replace the Newton iteration and accelerate optimal power flow [Donon et al., 2020, Zamzam & Baker, 2020], and sensitivity-informed training keeps learned outputs feasible [Nellikkath & Chatzivasileiadis, 2022]. Formal verification of learned grid outputs [Venzke & Chatzivasileiadis, 2021] and conformal prediction [Vovk et al., 2005, Lei et al., 2018] supply the guarantee machinery we use rather than invent, and neural operators [Li et al., 2021, Lu et al., 2021] motivate treating the queue as a batch of requests rather than a sequence of optimisations. The contribution here is to combine a screen with a guarantee at the queue level, where the cost being reduced is the number of security studies rather than the cost of one.

5 Limitations

The networks are academic, at 30 and 118 buses, and the two runs do not share a rating rule: case30 keeps its own conductor ratings while case118 has ratings synthesised at 0.55 of each branch's worst unperturbed post-contingency flow. That confounds the one cross-network comparison the paper makes, because the sign of the linear model's error could be a property of meshing or of the two rating rules, and this study cannot separate them; ratings and voltage bands also come from each network's own reference sweep, so the capacities are self-consistent but not numbers a utility would recognise.

The label is a verified lower bound at a finite tolerance, so the coverage figures are conservative in a way measured only for that tolerance, and the alternating-current model enforces no generator reactive limits. The shifted split is one demand level outside the training grid and 60.0% inadmissible, so its coverage figure rests on a small subset. The operator is trained on case30 only and the deferral rule is a rate rather than a schedule, so end-to-end clearing time is a projection; training on the pool's own output and a planning-grade network model are the next steps.

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Use of AI tools

Generative AI tools were used for review and polish rather than for drafting. Experimental design, baselines, metrics and interpretation were ours, and every number and claim was verified against stored artifacts: every value in the text is emitted as a macro from `tools/make_numbers.py` reading a results file, so none was transcribed by hand, and a macro whose source is absent prints as **??** on the page.

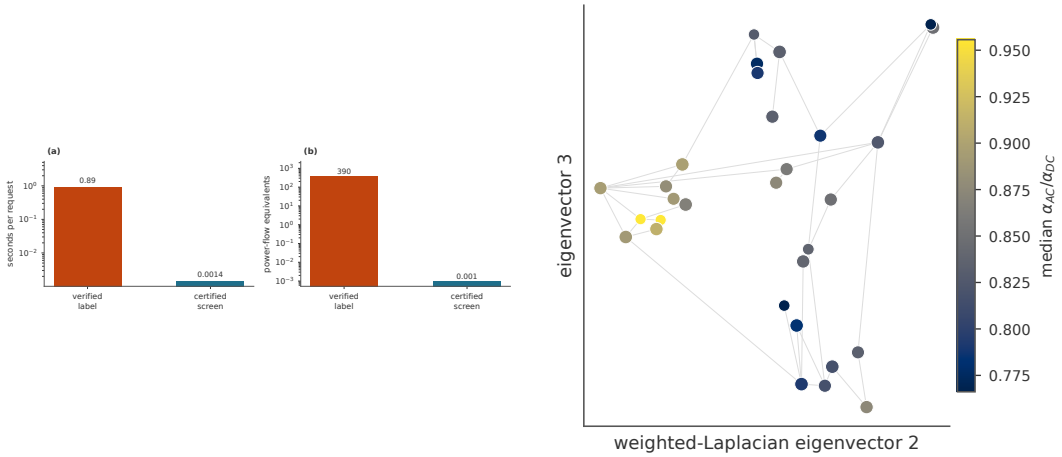


Figure 1: Left: cost per request, in seconds (machine dependent) and in power-flow equivalents (machine independent). The verified label is the baseline; the certified screen is the method. Right: median derate $\alpha_{\mathrm{AC}}/\alpha^{\star}$ per bus, over requests centred on that bus, on a layout of two weighted-Laplacian eigenvectors rather than a geographic map, marker size scaling with the number of requests that bus received. The colour is the size of the correction the learned step has to make to the linear answer, so the figure says where the linear model is wrong rather than where the network is congested.

Table 1: Equivalence of the pooled queue path to one-request-at-a-time bisection on case30, which is the claim the pooled path rests on. The baseline is serial bisection with an identical bracket, tolerance and verifier, over the 80 labels of a fixed run of states. "equal" means floating-point equality, so the equivalence covers the label, the stopping decision and the diagnostics together rather than only the capacity, and it holds at every queue length tested (4, 8, 16, 32requests). The cost comparison is in the main text and is a null result.

column	pooled vs one at a time
capacity α_{AC}	equal
binding mode	equal
censored and capped	equal
sweep evaluations	equal

A Splits, censoring and cost

Table 2: Per-split detail for the reported run on case30. The labelling mode is all studied outages per request over 41 branches, at a relative bisection tolerance of 5e-03, produced by the pooled path. "inadmissible" is the fraction of drawn operating states that fail their own base case before any request is added, so no request in them has a positive label. Coverage is at $1-\epsilon = 10\%$; the linear column is the rate at which the closed form exceeds the verified capacity.

split	requests	inadmissible	coverage	linear $>$ verified	censored
train (not scored)	869	9.2%	—	—	—
calibration	318	0.0%	—	—	—
test	311	2.5%	91.0%	95.5%	7
shifted demand	95	60.0%	66.3%	95.8%	—

The large network. The same pipeline is run on 118-bus case118, where a label costs a screened N-1 bisection over 177 outages. The screening is 12 outages chosen by the linear layer's own ranking, the label mode is screen, and the resulting study has 102 test requests over 186 branches. It is reported at that size rather than pooled with the small network because the two differ in the label budget they can afford, and pooling them would hide which of the two is carrying a result. The measured capacity the screen forgoes on a state sample is 1.00 of the all-outage capacity.

Table 3: Cost and conservativeness. The server-side cost of a verified label is reported in power-flow equivalents, which do not depend on the machine, next to the wall clock that does. "conservativeness" is the median ratio of the returned capacity to the verified capacity, so a value below one is capacity left on the table. Calibration uses only the 318 uncensored calibration requests out of 318.

quantity	value
verified label, median seconds	0.89
verified label, mean seconds	0.92
verified label, power-flow equivalents	390
certified screen, milliseconds per request	1.40
screen speedup, in power-flow equivalents	390×
screen speedup, in wall clock	636×
queue pooling speedup, 16-request check	2.18×
queue pooling speedup, over 80 labels	1.01×
certificate conservativeness, median	0.90
certificate conservativeness, 10th percentile	0.84
linear closed form conservativeness, median	1.20
point-prediction violation rate, test	47.9%
certificate violation rate, test / shifted	9.0% / 33.7%
binding-mode accuracy / balanced accuracy	100.0% / 100.0%
test requests by true mode (thermal / voltage / insecure)	1600 / 0 / 0
operator parameters, best epoch, best held-out loss	183428 / 26 / 0.00344

The large run has 486 labels over 115 states, of which 129 (26.5%) are censored, meaning the bracket top was still verified secure so the label is a bound rather than a measurement. Calibration uses the 79 uncensored requests out of 102. The certificate attains 94.1% coverage on the test split and 100.0% on the six admissible shifted requests, whose count is small enough that the second number is reported for completeness rather than relied on. The point prediction alone violates at 68.6%, so the certificate is doing real work here rather than restating an already accurate prediction; the certified capacity is 0.61 of the verified capacity at the median, against 0.46 for the linear closed form, and the certificate itself violates at 5.9%. A label costs 3.17 s at the median over 130 power-flow equivalents, against which the screen is 130× cheaper.

B What was checked, and against what

Table 4: Verification of the components the results depend on. Each row is produced by a committed tool and written to a JSON artifact, so a rerun updates the value rather than requiring the text to be edited. The solver cross-check is against PANDAPOWER; the shift-factor cross-check is against its `makePTDF`; the closed-form row compares Eq. 1 against a dense scan over directions, which is what makes "exact maximiser" a measurement rather than an assertion.

check	case	quantity
AC solver vs pandapower, voltage	case30	1.3e-15 pu max abs
AC solver vs pandapower, branch flow	case30	12.3% max rel
shift factors vs pandapower `makePTDF`	case14, case30, case57, protocol	2.7e-15 max abs
closed form vs dense scan	case14, case30, case57, protocol	7.4e-07 max rel
batched sweep vs serial solver	case30	covered by `feascap/tests`
pooled bisection vs one-at-a-time	case30	0% max rel
BLAS thread default vs pinned, stacked solve	178x181	0.78 s vs 0.622 s
BLAS thread factor	178x181	1.3×

The binding-limit head. The clearest negative result in the study. On case30 the test support is 1600 thermal against 0 voltage, and on case118 it is 373 thermal against 113 already-inadmissible, also with no voltage cases. So in neither study does a single request bind on voltage, and the reported accuracy of 100.0% is a one-class number that a constant predictor would also obtain. Two things follow and both are worth stating outright. The thermal rating rule sets every branch at a fraction of its own worst post-contingency flow, which leaves voltage as the looser constraint by construction, so this is a property of the study's limit definition as much as of the networks. And a three-class head is the wrong object for this data: what a queue operator needs is the binding outage and branch, which the operator already carries in its sensitivity features, not a limit category that is constant here.

Differentiable co-design. The operator is differentiable in the network it was built on, so the gradient of admitted capacity with respect to a candidate reinforcement is available without a second solve. On 3 test states the top reinforcement the gradient named was the one the exact alternating-current check confirmed as best in 2/3 cases. That sample is too small to support a claim about ranking quality, and it is reported as a demonstration that the gradient exists and points somewhere verifiable, not as evidence that it always points at the right line.

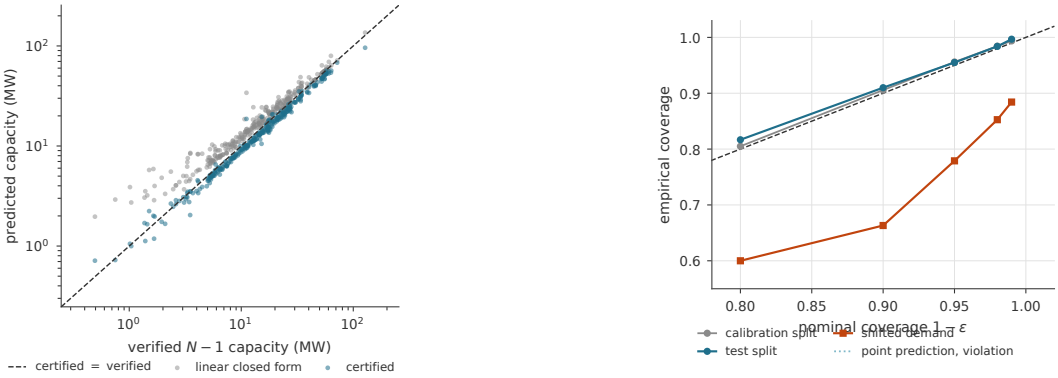


Figure 2: Left: predicted against verified $N-1$ capacity on 311 held-out test requests, log scale, dashed at equality. The grey cloud is the linear closed form, which sits above the diagonal and is the failure this paper is about; the accent cloud is the certified capacity, which sits below it. Right: empirical against nominal coverage for the calibration split (baseline), the test split and the shifted split. Markers are proportions over 311 and 95 requests, so the dispersion is the binomial standard error of 1.6% and 4.8% reported in the text.